
CUSTOMER CHURN ANALYSIS

Telco Customer Base — Risk Segmentation & Retention Strategy

26.5% churn rate , driven primarily by contract type and early-tenure risk

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Tools used :- Python(Pandas, Numpy, seaborn), Power BI

***Dataset :- WA_Fn-UseC_-Telco-Customer-Churn.csv | 7,043
Customers | 21 Fields***

Report generated July 2026

Executive Summary

This report analyzes churn behavior across 7,043 telecom customers to identify the strongest drivers of customer attrition, build a data-driven risk-scoring model, and translate those findings into a segmented retention strategy with an ROI case for action.

Overall churn rate: 26.5% of customers in the dataset have churned.

Four factors stand out as the clearest churn drivers:

- Contract type — month-to-month customers churn at 42.7%, vs. 11.3% (one-year) and 2.8% (two-year).
- Tenure — customers in their first 6 months churn at 53.3%, falling steadily to 9.5% past 48 months.
- Missing add-ons — customers without Tech Support or Online Security churn at roughly 3x the rate of those with it.
- Payment method & internet type — Electronic check payers (45.3%) and Fiber optic users (41.9%) churn well above other segments.

Using these drivers, a Risk Score and a Customer Value metric were built and combined into a 2x2 segmentation. The highest-priority group — High Risk / High Value customers (868 customers, ₹28.6 lakh in value, 44.9% churn) — is projected to lose ~390 customers without intervention. A targeted 10% retention discount to this segment is estimated to return 1.35x on cost even under conservative assumptions, making it a clear candidate for immediate action.

1. Dataset Overview

The dataset contains 7,043 customer records with 21 fields covering demographics, subscribed services, contract and billing details, and churn status. No missing values were found in the raw categorical fields; TotalCharges required conversion to numeric (11 blank values were coerced and filled with 0).

Metric	Value
Total customers	7,043
Total fields	21
Duplicate rows	0
Overall churn rate	26.5%
Average tenure	32.4 months
Average monthly charges	₹64.76

2. Exploratory Analysis: Churn Drivers

2.1 Churn Rate by Contract Type

Contract length is the single strongest churn driver in the dataset — a 15x gap separates the riskiest and safest contract types.

Contract Type	Churn Rate
Month-to-month	42.7%
One year	11.3%
Two year	2.8%

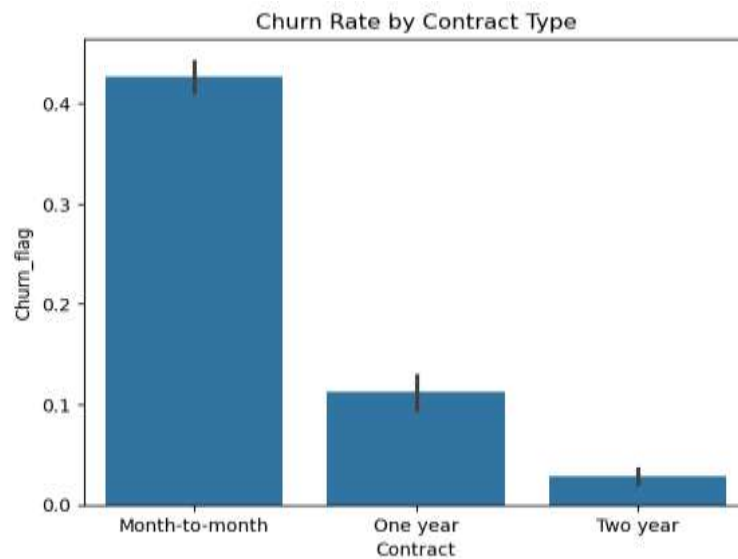


Figure 1. Churn rate by contract type.

2.2 Churn Rate by Tenure

Churn risk is heavily front-loaded: over half of customers in their first 6 months churn, compared to under 10% of customers past 4 years.

Tenure Group (months)	Churn Rate
0–6	53.3%
6–12	35.9%
12–24	28.7%
24–48	20.4%
48–72	9.5%

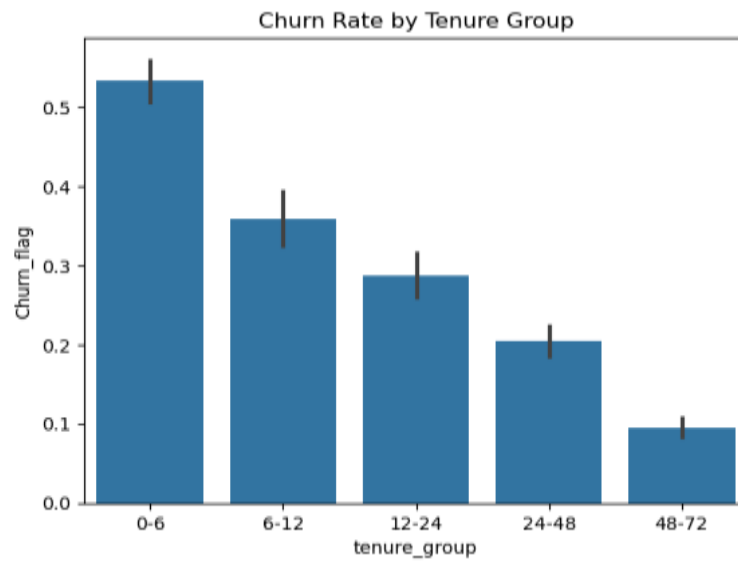
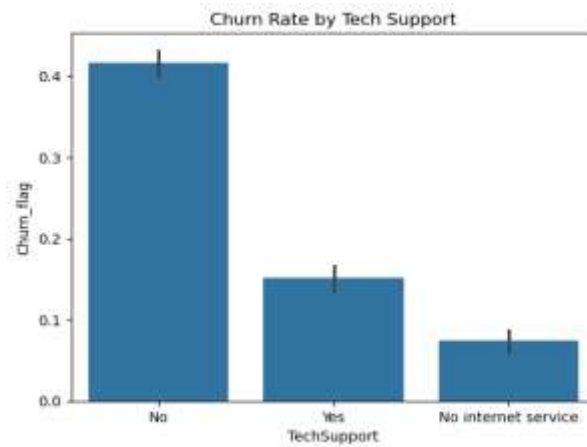


Figure 2. Churn rate by tenure group.

2.3 Churn Rate by Support & Security Add-ons

Customers lacking Tech Support or Online Security churn at close to 3x the rate of those who have these add-ons, suggesting the add-ons raise switching costs or improve customer experience.

Segment	Has Add-on	No Add-on	No Internet Service
Tech Support	15.2%	41.6%	7.4%
Online Security	14.6%	41.8%	7.4%



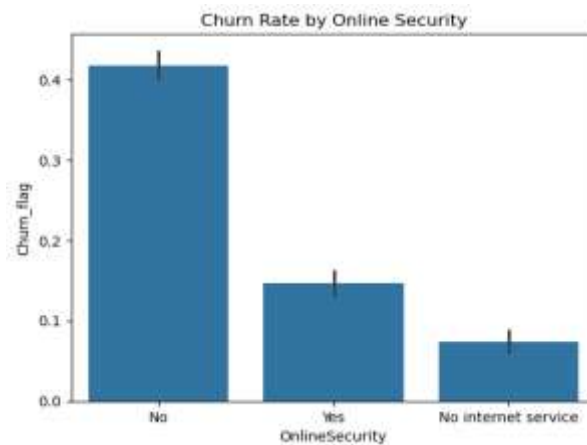
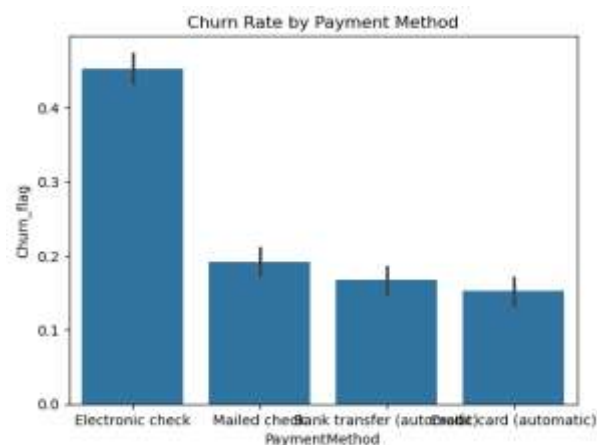


Figure 3–4. Churn rate by Tech Support and Online Security status.

2.4 Churn Rate by Payment Method & Internet Service

Electronic check payers and Fiber optic subscribers are two standout high-risk segments — notably, Fiber optic is the premium internet product, which points to possible pricing or service-quality dissatisfaction rather than product downgrade risk.

Payment Method	Churn Rate
Electronic check	45.3%
Mailed check	19.1%
Bank transfer (automatic)	16.7%
Credit card (automatic)	15.2%
Internet Service	Churn Rate
Fiber optic	41.9%
DSL	19.0%
No internet	7.4%



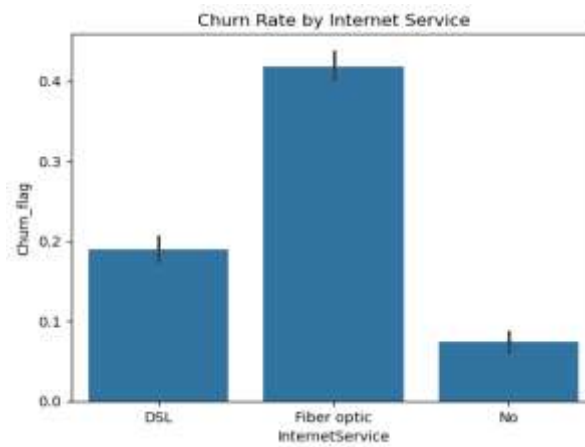


Figure 5–6. Churn rate by payment method and internet service type.

3. Risk Scoring Model

A composite Risk Score was built from the drivers identified above, weighting each factor by its observed churn impact:

- Month-to-month contract: +3
- Tenure under 6 months: +3
- Electronic check payment: +2
- No Tech Support: +2
- No Online Security: +2
- Fiber optic internet: +1

Scores were grouped into three risk tiers, which cleanly separate churn outcomes:

Risk Tier	Score Range	Churn Rate
Low	0–3	5.0%
Medium	4–7	21.2%
High	8–13	55.3%

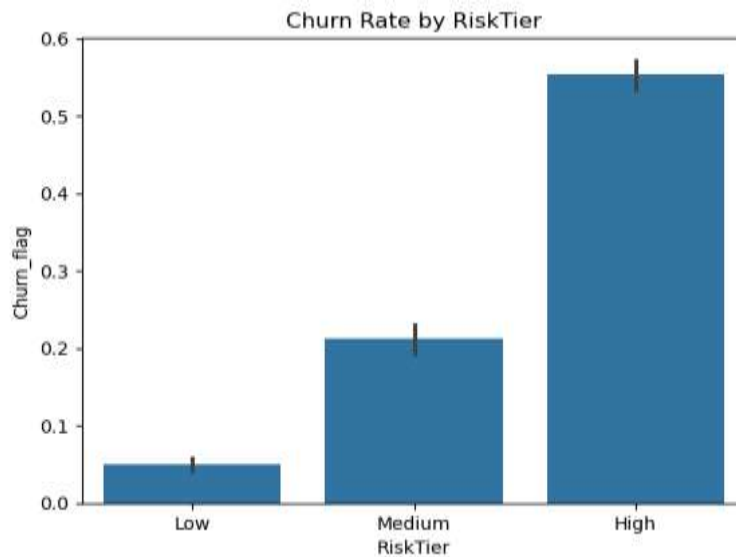


Figure 7. Churn rate by risk tier — an 11x spread between Low and High risk.

4. Customer Value Segmentation

A Customer Value metric (Monthly Charges × Tenure) was combined with Risk Tier to build a 2x2 segmentation, prioritizing where retention spend will have the greatest revenue impact.

Risk Tier	Value Tier	Customers	Avg. Value (₹)	Churn Rate
Low	High Value	1,696	4,476	6.0%
Low	Low Value	1,145	672	3.5%
Medium	High Value	957	4,000	17.8%
Medium	Low Value	793	507	25.3%
High	High Value	868	3,292	44.9%
High	Low Value	1,584	384	61.0%

5. Business Recommendation: ROI on Retention Offer

The High Risk / High Value segment — 868 customers, ~₹28.6 lakh in total value, 44.9% churn rate — is the highest-priority group for retention spend. Without intervention, an estimated 390 of these customers will churn, putting ~₹12.8 lakh in revenue at risk.

Metric	Value
Segment size	868 customers
Total segment value	₹28.6 lakh
Projected churners (no action)	~390 customers (~₹12.8L at risk)

Metric	Value
Retention discount cost (10%)	~₹2.86 lakh
Customers saved (30% success rate)	~117 customers
Revenue protected	~₹3.85 lakh
Return on campaign spend	1.35x

Even under a conservative 30% success assumption, the campaign returns more than it costs — making the offer clearly worth pursuing for this segment.

6. Retention Strategy by Segment

High Risk / High Value — 868 customers, 44.9% churn

Top priority. Proactive personal outreach — a retention call or account manager check-in rather than an automated email — paired with a targeted discount or free service upgrade (e.g., a free Tech Support add-on, shown to reduce churn ~3x). The ROI case above supports this spend.

High Risk / Low Value — 1,584 customers, 61.0% churn

Largest and riskiest group, but lowest value — not worth expensive interventions. Use low-cost, automated tactics: an email nudge, a small self-service discount code, or a prompt to move off Electronic Check to automatic payment (reduces involuntary churn from failed payments). Do not allocate phone or agent time here.

Medium Risk / High Value — 957 customers, 17.8% churn

Worth close monitoring rather than immediate action. A good candidate for an annual-contract upgrade incentive — offer a modest discount to switch to a 1-year plan, locking in both lower churn risk and predictable revenue.

Medium Risk / Low Value — 793 customers, 25.3% churn

Low-touch, low-cost tactics only — automated check-in emails, self-service account tips. Not worth manual outreach given the value size, but cheap to include in a broader retention email campaign.

Low Risk / High Value — 1,696 customers, 6.0% churn

Protect, don't chase. These are the best customers — the risk here isn't churn, it's complacency. Focus on loyalty and rewards rather than discounts (e.g., early feature access, small appreciation perks) to maintain satisfaction without unnecessary discount spend.

Low Risk / Low Value — 1,145 customers, 3.5% churn

Deprioritize entirely. Churn risk is already low and value is low, so this segment isn't worth active management right now.

7. Conclusion

Contract type, tenure, missing support add-ons, payment method, and internet service type are the clearest drivers of churn in this customer base. The Risk Score and Value segmentation built from these drivers point to a clear, ROI-backed priority order for retention spend — concentrating high-touch effort on the 868 High Risk / High Value customers first, while managing lower-value segments through low-cost, automated tactics.

Appendix: Interactive Dashboard

The analysis above is also available as an interactive dashboard, allowing filtering by gender, contract type, risk tier, internet service, and payment method.

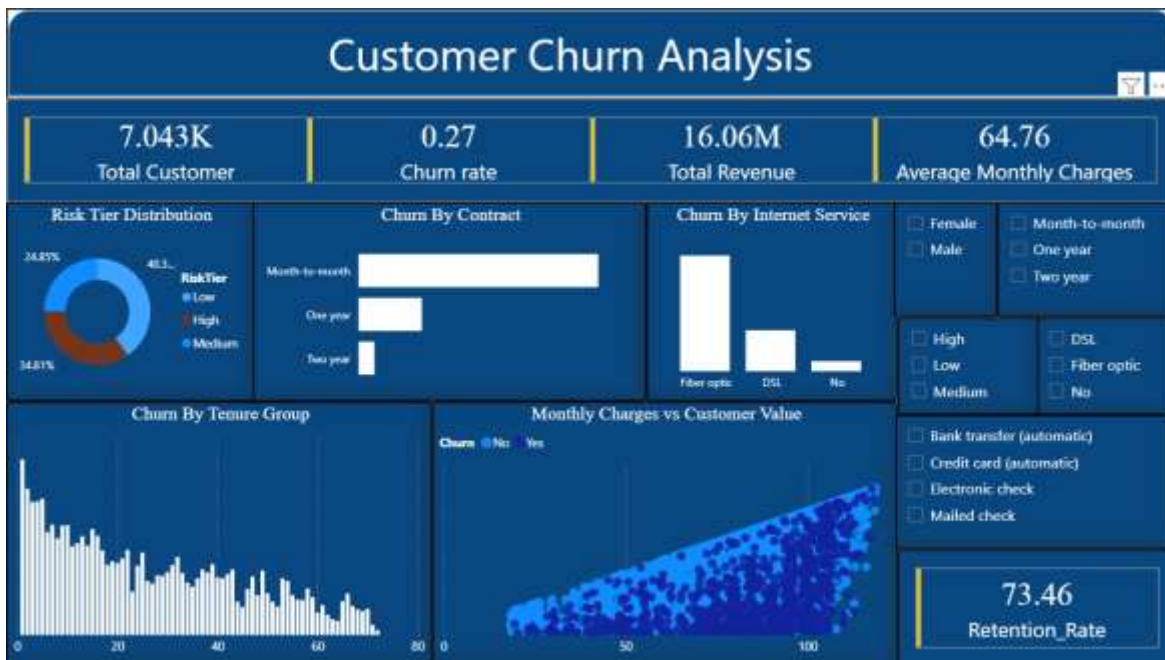


Figure 8. Customer Churn Analysis dashboard — KPIs, risk tier distribution, churn by contract/tenure/internet service, and monthly charges vs. customer value.